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SARIMA Model Simulator

Simulation, differencing, and ACF/PACF analysis

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1 Introduction

The simulator allows users to build and visualize time series generated from SARIMA models of the form:

$$SARIMA(p, d, q)(P, D, Q)_{12}$$

Its purpose is not only to simulate a time series, but also to support the understanding of the main components of a seasonal ARIMA model:

- non-seasonal autoregressive dependence;
- non-seasonal moving average dependence;
- the presence of seasonal structure;
- the effect of ordinary and seasonal differencing;
- the interpretation of the ACF and PACF;
- the relationship between model parameters and the patterns observed in correlograms.

This guide is designed to promote active learning through systematic experimentation with the model orders and their coefficients.

2 Functional structure of the simulator

The interface is divided into two main areas.

2.1 Parameter panel

The side panel is used to configure the model orders:

$$(p, d, q)(P, D, Q)_{12}$$

where:

- p : non-seasonal autoregressive order;
- d : number of ordinary differences;
- q : non-seasonal moving average order;
- P : seasonal autoregressive order;
- D : number of seasonal differences;
- Q : seasonal moving average order.

The **Advanced** button allows users to modify the model coefficients:

$$(1 - \Phi_1 B^{12} - \dots - \Phi_P B^{12P})(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d(1 - B^{12})^D Y_t = (1 + \Theta_1 B^{12} + \dots + \Theta_Q B^{12Q})(1 + \theta_1 B + \dots + \theta_q B^q) \varepsilon_t$$

- ϕ_i : AR coefficients;
- θ_i : MA coefficients;
- Φ_i : seasonal AR coefficients;
- Θ_i : seasonal MA coefficients.

It also allows users to modify the length of the simulated series and the maximum number of lags displayed in the ACF/PACF.

2.2 Results panel

The main panel displays:

- the simulated time series;

- the differenced series;
- the ACF of the differenced series;
- the PACF of the differenced series;
- a table with the simulated data.

Note: the ACF and PACF are computed on the differenced series, not on the original series.

3 Interpretation of the plots

3.1 Simulated series

The first figure shows the series generated by the selected model. It allows users to visually inspect:

- trend;
- temporal persistence;
- seasonal oscillations;
- changes produced by the AR, MA, SAR, and SMA parameters.

3.2 Differenced series

The second figure shows the series after applying the selected differences. Its goal is to check whether the series becomes more stable after removing trend or seasonality.

3.3 ACF

The ACF measures the correlation between the series and its lags. In the simulator:

- grey bars do not exceed the reference intervals;
- blue bars indicate significant short lags;
- yellow bars indicate significant seasonal lags, such as 12 or 24.

The ACF is especially useful for detecting MA components and seasonal patterns.

3.4 PACF

The PACF measures the partial correlation between the series and its lags, removing the effect of intermediate lags.

The PACF is especially useful for detecting AR components.

4 Guided example

4.1 Scenario 1: AR(1) model

Select:

$$SARIMA(1, 0, 0)(0, 0, 0)_{12}$$

with $\phi_1 > 0$.

Observe that:

- the series shows persistence;
- the ACF decreases gradually;
- the PACF shows a prominent bar at lag 1.

Guiding question: what happens if you increase ϕ_1 towards values close to 1?

4.2 Scenario 2: MA(1) model

Select:

$$SARIMA(0, 0, 1)(0, 0, 0)_{12}$$

Observe that:

- the ACF shows a prominent bar at lag 1;
- the PACF decreases progressively.

Guiding question: why does the main pattern now appear in the ACF rather than in the PACF?

4.3 Scenario 3: seasonal model

Select:

$$SARIMA(0, 0, 0)(1, 0, 0)_{12}$$

Observe that:

- relevant bars appear at seasonal lags;
- lags 12, 24, or 36 may stand out in the ACF/PACF.

Guiding question: what does a significant bar at lag 12 indicate?

5 Fundamental concepts illustrated by the simulator

AR process. An autoregressive process uses past values of the series itself to explain the current value. The PACF usually provides a particularly informative structure for identifying its order.

MA process. A moving average process uses past errors to explain the current value. The ACF usually provides a particularly informative structure for identifying its order.

Seasonality. Seasonality occurs when the series shows dependence at lags separated by a fixed period, for example 12 in monthly data.

Ordinary differencing. Ordinary differencing is used to remove trend or long-term dependence by taking differences between consecutive observations.

Seasonal differencing. Seasonal differencing is used to remove patterns that repeat every s periods, for example every 12 months.

ACF. The ACF measures the correlation between the series and its lags. It is useful for studying temporal dependence and detecting possible MA or seasonal terms.

PACF. The PACF measures the partial correlation with each lag, discounting the information explained by intermediate lags. It is useful for studying possible AR terms.

6 How to study with the simulator

An effective approach is:

1. Start with a simple model.
2. Change only one parameter at a time.
3. Observe how the series, ACF, and PACF change.
4. Compare AR, MA, SAR, and SMA models.
5. Introduce ordinary or seasonal differencing and observe its effect.

6. Try to infer the model by looking only at the ACF and PACF.

In this way, the simulator works as a laboratory for understanding how SARIMA models generate observable temporal patterns.